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United States Patent	12410944
Kind Code	B2
Date of Patent	September 09, 2025
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Water heating system

Abstract

A water heating system having a tank and one or more heat sources for heating water is provided. The water heating system includes a water outlet for allowing egress of water from the tank and a dip tube for allowing ingress of water into the tank. The dip tube includes a first end for coupling with a water source and a second end disposed proximate a base of the tank for discharging water into the tank through an array of holes. Each of the holes is defined in a side wall of the dip tube and laterally discharges water with respect to a longitudinal axis of the second end of the dip tube.

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Appl. No.: 17/547555

Filed: December 10, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220186981 A1	Jun. 16, 2022

Related U.S. Application Data

us-provisional-application US 63124180 20201211

Publication Classification

Int. Cl.: F22D7/04 (20060101); F24H1/18 (20220101); F24H9/13 (20220101); F24H9/14 (20060101)

U.S. Cl.:

CPC F24H9/133 (20220101); F24H1/18 (20130101);

Field of Classification Search

CPC: F22D (7/00); F22D (7/04); F24H (9/0042); F24H (9/14)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit, under 35 U.S.C. § 119, of U.S. Provisional Patent Application No. 63/124,180, filed 11 Dec. 2020, the entire contents and substance of which is incorporated herein by reference in its entirety as if fully set forth below.

TECHNICAL FIELD

(1) The present disclosure relates, in general, to a water heater and, more specifically relates, to a dip tube used for receiving water into a tank of the water heater.

BACKGROUND

(2) Water heaters are used to heat and store a quantity of water in a tank thereof for subsequent, on-demand delivery of hot water for residential and commercial use. Electric water heaters utilize heating elements to transfer heat to the water contained in the tank and can be controlled by thermostat devices that monitor temperature of the water in the tank. Further, the water heater includes a dip tube for receiving water from an external source into the tank and a bottom end of the dip tube is positioned at a bottom of the tank.

(3) Generally, it is beneficial to maintain a uniform temperature of water within the tank. However, the water at the bottom of the tank tends to be at a lower temperature as the water from an external source is generally cooler when discharged at the bottom of the tank. The temperature rises as the level of water approaches a top of the tank. In a typical water heating system, the stream of colder water discharged from the bottom end of the dip tube may have higher momentum, and thereby the colder water may move upward at a greater velocity towards the warmer water present at the top of the tank. As a result, the colder water will mix with the hot water, and the overall temperature of the water at higher levels of the tank (near outlet) may get impacted (reduced at an undesirably fast rate). Relatedly, a first-hour rating (FHR) of the water heater, which is a measure of a volume of hot water that the water heater can supply in the first hour of operation, may be negatively impacted. The FHR is an industry-wide indicator (as measured by a standardized test) used to establish thermal efficiency of the water heater, and hence water heater manufacturers continually strive to increase the first-hour rating. Under the FHR test, water draws continue until the water outlet temperature of the tank decreases by 15 degree F. from its maximum value, which is typically observed within the first 30 seconds of that the initial draw.

SUMMARY

(4) According to an aspect of the present disclosure, a water heating system is disclosed. The water heating system includes a tank and one or more heat sources in thermal communication with the tank. The one or more heat sources are configured to heat water within the tank. The water heating system further includes a water outlet configured to allow egress of water from the tank. The water heating system further includes a dip tube configured to allow ingress of water into the tank. The dip tube includes a first end configured to couple with a water source and a second end distal to the first end. The second end of the dip tube is disposed proximate a base of the tank and configured to discharge water to the tank through an array of holes. Each of the holes is defined in a side wall of the dip tube and is configured to laterally discharge water with respect to a longitudinal axis of the dip tube. The second end of the dip tube can be closed. A sum of cross-sectional area of each hole of the array of holes is greater than a cross-sectional area of the second end of the dip tube.

Particularly, the sum of cross-sectional area of each hole of the array of holes is at least 50 percent higher than the cross-sectional area of the second end of the dip tube. The water heating system can further include one or more guide members disposed at an inner surface of the side wall of the dip tube and configured to guide discharge of the water through the array of holes.

(5) In one embodiment, the array of holes includes one or more rows of holes radially defined around the longitudinal axis of the second end of the dip tube. Each row contains at least two holes defined at equilateral distance. In another embodiment, the array of holes includes a first row of holes proximate the second end of the dip tube and a second row of holes defined adjacent the first

row of holes. Each hole of the second row is aligned at an approximate middle of two adjacent holes of the first row. In yet another embodiment, the array of holes includes one or more rows of holes radially defined around the longitudinal axis of the second end of the dip tube. Each row includes two or more holes defined with respect to a side portion of the side wall of the dip tube that does not contain any holes.

(6) In an embodiment, the dip tube includes a body portion and an elongated portion defining the array of holes and the second end. The elongated portion is detachably coupled with the body portion of the dip tube. Each hole of the array of holes defined in the elongated portion is positioned to direct the water towards the wall of the tank. Further, each hole of the array of holes defined in the elongated portion is adapted to direct the water substantially along the horizontal plane. In one embodiment, the body portion is coaxially and vertically aligned with respect to the longitudinal axis of the second end of the dip tube. In another embodiment, the body portion is aligned perpendicular with respect to the longitudinal axis of the second end of the dip tube and parallel to the base of the tank. In yet another embodiment, a cross sectional area of the elongated portion tapers towards the base of the tank. In one embodiment, a cross sectional shape of each of the holes is a circle. In another embodiment, a cross sectional shape of each of the holes is a rectangle.

(7) These and other aspects and features of non-limiting embodiments of the present disclosure will become apparent to those skilled in the art upon review of the following description of specific non-limiting embodiments of the disclosure in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A better understanding of embodiments of the present disclosure (including alternatives and/or variations thereof) may be obtained with reference to the detailed description of the embodiments along with the following drawings, in which:

(2) FIG. 1 is a cross-sectional view of a water heating system, according to an embodiment of the present disclosure;

(3) FIG. 2 is an enlarged view of a portion 'C' of a dip tube of the water heating system shown in FIG. 1, according to an embodiment of the present disclosure;

(4) FIG. 3 is an enlarged view of a second end of a dip tube having an array of holes, according to one embodiment of the present disclosure;

(5) FIG. 4 is an enlarged view of a second end of a dip tube having an array of slots, according to another embodiment of the present disclosure;

(6) FIG. 5 is an enlarged view of a second end of a dip tube having an array of holes, according to yet another embodiment of the present disclosure;

(7) FIG. 6A is an enlarged view of a second end of a dip tube having a vertically disposed elongated portion, according to an embodiment of the present disclosure;

(8) FIG. 6B is an enlarged view of a second end of a dip tube having a laterally disposed elongated portion, according to an embodiment of the present disclosure; and

(9) FIG. 6C is an enlarged view of a second end of a dip tube having a tapered elongated portion, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) Throughout this disclosure, various aspects of the disclosed technology can be presented in a range format (e.g., a range of values). It should be understood that such descriptions are merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the disclosed technology. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual rational numerical values

within that range. For example, a range described as being “from 1 to 6” includes the values 1, 6, and all values therebetween. Likewise, a range described as being “between 1 and 6” includes the values 1, 6, and all values therebetween. The same premise applies to any other language describing a range of values. That is to say, the ranges disclosed herein are inclusive of the respective endpoints, unless otherwise indicated.

(11) Herein, the use of terms such as “having,” “has,” “including,” or “includes” are open-ended and are intended to have the same meaning as terms such as “comprising” or “comprises” and not preclude the presence of other structure, material, or acts. Similarly, though the use of terms such as “can” or “may” are intended to be open-ended and to reflect that structure, material, or acts are not necessary, the failure to use such terms is not intended to reflect that structure, material, or acts are essential. To the extent that structure, material, or acts are presently considered to be essential, they are identified as such.

(12) In the following description, numerous specific details are set forth. But it is to be understood that embodiments of the disclosed technology may be practiced without these specific details. In other instances, well-known methods, structures, and techniques have not been shown in detail in order not to obscure an understanding of this description. References to “one embodiment,” “an embodiment,” “example embodiment,” “some embodiments,” “certain embodiments,” “various embodiments,” etc., indicate that the embodiment(s) of the disclosed technology so described may include a particular feature, structure, or characteristic, but not every embodiment necessarily includes the particular feature, structure, or characteristic. Further, repeated use of the phrase “in one embodiment” does not necessarily refer to the same embodiment, although it may.

(13) Throughout the specification and the claims, the following terms take at least the meanings explicitly associated herein, unless the context clearly dictates otherwise. The term “or” is intended to mean an inclusive “or.” Further, the terms “a,” “an,” and “the” are intended to mean one or more unless specified otherwise or clear from the context to be directed to a singular form.

(14) Unless otherwise specified, the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicate that different instances of like objects are being referred to and are not intended to imply that the objects so described should be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

(15) Reference will now be made in detail to example embodiments of the disclosed technology, examples of which are illustrated in the accompanying drawings and disclosed herein. Wherever convenient, the same reference numbers will be used throughout the drawings to refer to the same or like parts. Moreover, references to various elements described herein, are made collectively or individually when there may be more than one element of the same type. However, such references are merely exemplary in nature. It may be noted that any reference to elements in the singular may also be construed to relate to the plural and vice-versa without limiting the scope of the disclosure to the exact number or type of such elements unless set forth explicitly in the appended claims.

(16) Referring to FIG. 1, a cross-sectional view of a water heating system **100** is illustrated, according to an embodiment of the present disclosure. The water heating system **100** (hereinafter referred to as ‘the system **100**’) includes a tank **102** having a wall **104**. In an example, capacity of the system **100** may be, but is not limited to, 30, 40, 50, 55, 60, 75, 80, 85, or 100 gallons. The system **100** also includes a dip tube **106** and a water outlet **108** defined through the wall **104**. An inlet fitting **110** and an outlet fitting **112** are provided in the wall **104**, at a top end **114** of the tank **102**, to respectively support the dip tube **106** and the water outlet **108** routed through the wall **104**. The dip tube **106** is configured to allow ingress of water into the tank **102** and the water outlet **108** is configured to allow egress of water from the tank **102**. Heated water is drawn from the tank **102** through the water outlet **108** with aid of, for example, a pump, and may be delivered to one or more end devices, such as laundry washers, dishwashers, faucets, and shower heads. In an embodiment, the wall **104** of the tank **102** may be insulated to retain heat of the water for longer duration.

(17) The system **100** further includes one or more heat sources in thermal communication with the

tank. The one or more heat sources can be or include any source of heat capable of heating water (e.g., one or more electrical heating elements, combustion-type heating, heat pump). As illustrated, the system can include one or more heating elements coupled to the wall **104** and configured to heat water. In an embodiment, a first heating element **116** is disposed proximate the top end **114** of the tank **102** and a second heating element **118** is disposed proximate a base **120** of the tank **102**. The first heating element **116** and the second heating element **118** may be attached to the wall **104** using a first coupling **124** and a second coupling **126**, respectively. Electric current supply to the first heating element **116** and the second heating element **118** may be routed through the first coupling **124** and the second coupling **126**, respectively. The system **100** can further include a first thermostat **128** and a second thermostat **130** configured to sense temperature of water proximate the top end **114** and the base **120**, respectively, of the tank **102**. The first heating element **116** and the second heating element **118** can extend parallel to the base **120**.

(18) The dip tube **106** extends from the inlet fitting **110** downward towards the base **120** of the tank **102** to supply colder water to the tank **102**. The dip tube **106** includes a first end **132** configured to couple with a water source **130** and a second end **134** distal to the first end **132**. The second end **134** of the dip tube **106** is disposed proximate the base **120** of the tank **102**, i.e., closer to the base **120** than to the top end **114** of the tank **102**. The dip tube **106** is typically positioned adjacent to the wall **104** of the tank **102**. The dip tube **106** may have a length defined between the first end **132** and the second end **134**. In one embodiment, the length of the dip tube **106** may be in a range of about 50 inches to about 60 inches. In various embodiments, the length of the dip tube **106** may be defined based on various factors including, but not limited to, a height of the tank **102** and capacity of the system **100**. In an embodiment, a cross-sectional shape of the dip tube **106** may be a circle. In some embodiments, the cross-sectional shape of the dip tube **106** may be an oval, an ellipse, or any polygon shape known in the art.

(19) As the base **120** may develop scales over a period of use of the system **100**, the second end **134** of the dip tube **106** is located at a predetermined height from the base **120** of the tank **102** to prevent being subjected to scaling or prevent any influence of such scaling on the intended function of the dip tube **106**. The predetermined height of the dip tube **106** from the base **120** of the tank **102** may be defined based on various factors including, but not limited to, type of material, size and cross-sectional shape of the dip tube **106**.

(20) The dip tube **106** further includes an array of holes **140** defined at the second end **134** thereof and configured to discharge water to the tank **102** therethrough. Each of the holes **140** is defined in a side wall **142** of the dip tube **106** and the second end **134** of the dip tube **106** is closed. That is, the terminal, axial opening defined by the circumference of the dip tube **106** can be closed or sealed (e.g., end wall **144**, shown in FIG. 2), while the side wall **142** contains a plurality of apertures **140** through which the water is discharged from the dip tube **106**.

(21) The array of holes **140** is configured to laterally discharge the water with respect to a longitudinal axis 'L' (shown in FIG. 2) of the second end **134** of the dip tube **106**. In an embodiment, the array of holes **140** may include one or more rows of holes radially defined around the longitudinal axis 'L' of the dip tube **102**, and each row may include at least two holes. Each hole of the array of holes **140** defined at the second end **134** of the dip tube **106** is positioned to direct the water towards the wall **104** of the tank **102**. Particularly, the array of holes **140** defined at the second end **134** of the dip tube **106** is adapted to direct water substantially along a horizontal plane. The array of holes **140** may be defined in a desired pattern in the side wall **142** of the dip tube **106**. The various embodiments of the desired patterns of the holes **140** will be described in detail herein below. The array of holes **140** defined in the side wall **142** may be otherwise referred to as lateral openings of the dip tube **106** and configured to discharge the water laterally within the tank **102** while restricting discharge of the water axially. With the array of holes **140**, the water may discharge slowly and in a more horizontal and diffused manner compared to a vertical discharge of the water through an axial opening of the dip tube, in which the stream of colder water may achieve

a higher momentum and may move upward at greater velocity towards the water outlet **108** after bouncing back from the base **120** of the tank **102**.

(22) In some embodiments, a sum of the cross-sectional area of each hole of the array of holes **140** is greater than a cross-sectional area of the dip tube **106** at the second end **134**. As such, a larger quantity of water is discharged through the array of holes **140** substantially along a horizontal plane compared to a vertical discharge of the water through an axial opening of the dip tube **106**. Further, the second end **134** of the dip tube **106** is terminally closed at end wall **144** to avoid discharging of the water along a longitudinal plane or vertically towards the base **120** of the tank **102**. In another embodiment, the sum of the cross-sectional area of each hole of the array of holes **140** may be 50 percent higher than the cross-sectional area of the dip tube **106** at the second end **134**. In one example, the sum of the cross-sectional area of each hole of the array of holes **140** may be at least double the cross-sectional area of the dip tube **106** at the second end **134**. In another example, a ratio between the sum of the cross-sectional area of each hole of the array of holes **140** and the cross-sectional area of the dip tube **106** at the second end **134** may be in a range of about 4 to 6.

(23) During operation, colder water is supplied into the tank **102** through the dip tube **106**. The water discharges through the array of holes **140** defined at the second end **134** of the dip tube **106** laterally and in a diffused manner towards the base **120** of the tank **102**. Due to the manner in which the array of holes **140** is arranged, the velocity of the water exiting the dip tube **106** reduces and the water flows generally horizontally in order to prevent the streams of colder water from mixing into the warm water present at the top end **114** of the tank **102**. Each draw during the first hour rating (FHR) test ends when the temperature of water at the water outlet **108** drops by 15 degree F. from its maximum value, which is typically observed in the first 30 s of that the initial draw. With the dip tube **106** of the present disclosure, the velocity of colder water exiting the dip tube **106** is reduced (e.g., as compared to the traditional dip tubes with an open bottom end and no side holes). As a result, the cold water gets mixed with the existing hot water of the tank more slowly, and the temperature of the water at the water outlet **108** decreases slowly, effectively slowing down the outlet temperature drop and thereby increasing the first draw volume capacity, which ultimately increases the FHR of the tank **102**.

(24) In an embodiment, sensors (e.g., temperature sensors) may be located within the tank **102** to sense temperature of water in the tank **102**. Accordingly, based on the water temperature, each of the first heating element **116** and the second heating element **118** may be operated to heat the water. Once the temperature of the water heated in the tank **102**, sensed by the first thermostat **128** or the second thermostat **130** (which can include or be in communication with the temperature sensors), reaches a predetermined value, the water is drawn from the tank **102**. With the dip tube **106** of the present disclosure, the first-hour rating (FHR) of the system **100** is increased by up to 10 percent or more, such as from about 1 percent to about 10 percent, as compared to known water heating systems (i.e., a system lacking the dip tube **106** and having a traditional dip tube with a terminal opening defined by the circumference of the second end of the dip tube). That is to say, the dip tube is configured to discharge the water such that a first hour rating of the water heater system with the dip tube is up to 10 percent greater than a first hour rating of the water heater system having a conventional dip tube.

(25) Referring to FIG. 2, an enlarged view of a portion 'C', specifically the second end **134** of the dip tube **106** shown in FIG. 1, is illustrated, according to an embodiment of the present disclosure. The dip tube **106** includes the side wall **142** and an end wall **144** defining a closed terminal end for the dip tube **106**, thereby discharging the water axially from the second end **134** of the dip tube **106** is avoided. The dip tube **106** further includes the array of holes **140** defined in the side wall **142**.

Each hole of the dip tube **106** is individually referred to as 'the hole **146**' and collectively referred to as 'the array of holes **140**' or 'the plurality of holes **140**' unless otherwise specifically mentioned.

(26) The array of holes **140** can include a plurality (three are illustrated in FIG. 2) of rows of holes radially defined around the longitudinal axis 'L' of the second end **134** dip tube **106**. Each row of

holes may be individually referred to as ‘the row of holes **148**’ or ‘the row **148**’. Each of the three rows **148** may be defined along a plane perpendicular to the longitudinal axis ‘L’ of the dip tube **106**. In some embodiments, the array of holes **140** may include two or more rows of holes radially defined around the longitudinal axis ‘L’ of the dip tube **106**. A distance between the two adjacent rows of the holes **148** may be defined based on various factors including, but not limited to, cross sectional area of the dip tube **106**, the length of the dip tube **106** and the capacity of the tank **102**. (27) Each row **148** can include a plurality (three are illustrated in FIG. 2) of holes **146** defined at equilateral distance. In the embodiment of FIG. 2, each of the three holes **146** of each row **148** is positioned 120 degrees apart from each other. A cross-sectional shape of the hole **146** is a circle, although other shapes may also be used, as described below.

(28) In certain embodiments, the diameter of the hole **146** may be scaled up or down proportionally based on the diameter of dip tube **106**. In an embodiment, the diameter of the dip tube **106** at the second end **134** may be in a range of about 18 mm to 22 mm. For a 19 mm diameter dip tube **106**, at least 12 mm diameter holes may be defined at the side wall **142**. Further, the diameter of the hole **146** may be in a range of about 60 to 70 percent of the diameter of the dip tube **106**. In another example, the diameter of the holes **146** may be in a range of about 12 mm to 15 mm.

(29) In an alternate embodiment, each row **148** may include two holes **146**, and each of the two holes **146** may be positioned 180 degrees apart from each other. In such a case, the diameter of the hole **146** may be in a range of about 15 mm to 20 mm. Further, the diameter of the hole **146** may be defined in a range of about 70 to 90 percent of the diameter of the dip tube **106**. In some embodiments, each row **148** may include four holes **146**, and each of the four holes **146** may be positioned 90 degrees apart from each other.

(30) In an alternate embodiment, the cross-sectional shape of the hole **146** may be a rectangle. In certain embodiments, the cross-sectional shape of the hole **146** may be an oval, an ellipse, or any polygon shape known in the art.

(31) In certain embodiments, as illustrated in FIG. 2, the array of holes **140** includes a first row of holes, alternatively referred to as ‘the first row **148-1**’ and a second row of holes, alternatively referred to as ‘the second row **148-2**’, defined adjacent the first row of holes. Particularly, the first row **148-1** is defined proximate the end wall **144** and the second row **148-2** is defined above the first row **148-1**. Each hole **146** of the second row **148-2** is aligned midway between two adjacent holes **146** of the first row **148-1**. In one example, each of the first row **148-1** and the second row **148-2** includes three holes **146** at 120 degrees apart, and each hole **146** of the second row **148-2** is offset by 60 degrees radially with respect to each hole **146** of the first row **148-1**. In another example, each of the first row **148-1** and the second row **148-2** may include two holes at 180 degrees apart, and each hole **146** of the second row **148-2** may be offset by 90 degrees radially with respect to each hole **146** of the first row **148-1**. In yet another example, each of the first row **148-1** and the second row **148-2** may include four holes **146** at 90 degrees apart, and each hole **146** of the second row **148-2** may be offset by 45 degrees radially with respect to each hole **146** of the first row **148-1**.

(32) Referring to FIG. 3, an enlarged view of a second end **234** of a dip tube **206** is illustrated, according to an embodiment of the present disclosure. The dip tube **206** includes a side wall **242** and an end wall **244** defining a closed terminal end for the dip tube **206**. The dip tube **206** further includes an array of holes **240** defined in the side wall **242**. The array of holes **240** includes three rows of holes **248** radially defined around a longitudinal axis of the dip tube **206**. Each of the three rows **248** may be defined along a plane perpendicular to the longitudinal axis of the dip tube **206**. Each row **248** includes four holes **246** defined at equilateral distance. Particularly, each of the four holes **246** is positioned 90 degrees apart from each other. Further, the holes **246** of each of the three rows **248** are aligned vertically and thereby define four individual columns of the holes **246**, and thus a symmetrical orientation of the hole **246** is achieved at the second end **234** of the dip tube **206**. A sum of cross-sectional area of each hole of the array of holes **240** is greater than a cross-

sectional area of the dip tube **106**. A cross-sectional shape of the hole **246** is a circle, and diameter of the dip tube **206** may be in a range of about 18 mm to 22 mm. In an example, for a 19 mm diameter dip tube **206**, at least 12 mm diameter holes **246** may be defined at the side wall **242**. In an alternative embodiment, the cross-sectional shape of the hole **246** may be a rectangle. In certain embodiments, the cross-sectional shape of the hole **246** may be an oval, an ellipse, or any polygon shape known in the art.

(33) A distance between the two adjacent rows **248** may be defined based on various factors including, but not limited to, cross sectional area of the dip tube **206**, a length of the dip tube **206** and the capacity of the tank **102**.

(34) In certain embodiments, the dip tube **206** further includes one or more guide members **250** disposed at an inner surface **252** of the side wall **242**. In various embodiments, the one or more guide members **250** may be oriented perpendicular, or at an inclination, with respect to the inner surface **252** of the side wall **242** to guide discharge of the water through the array of holes **240**. Particularly, the one or more guide members **250** may be provided at a bottom peripheral edge of the holes **246** of the row that is distal from the end wall **244** of the dip tube **206**. In an alternate embodiment, the one or more guide members **250** may be provided at a bottom peripheral edge of each hole **246** of each row **248**.

(35) Referring to FIG. 4, an enlarged view of a second end **334** of a dip tube **306** is illustrated, according to an embodiment of the present disclosure. The dip tube **306** includes a side wall **342** and an end wall **344** defining a closed terminal end for the dip tube **306**. The dip tube **306** further includes an array of slots **340** defined in the side wall **342**. The array of slots **340** includes one row of slots **348** radially defined around a longitudinal axis of the dip tube **306**. The row **348** may be defined along a plane perpendicular to the longitudinal axis of the dip tube **306** at the second end **334**. The row **348** includes multiple slots **346** and each of the multiple slots **346** may be defined at equilateral distance. In an example, the row **348** may include three to five slots **346**. Each slot **346** has a rectangular cross-sectional shape, in which a longer length of the slot **346** is aligned parallel to the longitudinal axis of the dip tube **306** at the second end **334**. A cross-sectional shape of the dip tube **306** is a circle. A sum of cross-sectional area of each hole of the array of holes **340** is greater than a cross-sectional area of the dip tube **106**. The dip tube **306** may further include a conical protrusion **350** extending inwardly from the end wall **344** (e.g., extending vertically upward). The conical protrusion **350** may have a height equal to or greater than the longer length of the slot **346**. The longer length and a width of each slot **346** may be defined based on various factors including, but not limited to, cross sectional area of the dip tube **306**, length of the dip tube **306** and the capacity of the tank **102**.

(36) Referring to FIG. 5, an enlarged view of a second end **434** of a dip tube **406** is illustrated, according to an embodiment of the present disclosure. The dip tube **406** includes a side wall **442** and an end wall **444** defining a closed terminal end for the dip tube **406**. The dip tube **406** further includes an array of holes **440** defined in the side wall **442**. The array of holes **440** includes three rows of holes **448** radially defined around a longitudinal axis of the second end **434** of the dip tube **406**. Each row **448** may be defined along a plane perpendicular to the longitudinal axis of the second end **434** of the dip tube **406**. Each row **448** includes three holes **446** defined with respect to a side portion **450** of the side wall **442** containing no holes. As such, the array of holes **440** define an asymmetric orientation of the holes **446**. A visual indicator (not shown) may be provided in the dip tube **406** to indicate the side portion **450** of the side wall **442** such that, during assembly of the system **100**, the dip tube **406** may be inserted into the tank **102** to face the side portion **450** of the side wall **442** to the closest inner surface portion of the tank **102**. That is to say, the dip tube **406** can be oriented such that no holes are located on the side of the dip tube **406** that is facing the closest inner surface portion of the tank **102**. Stated otherwise, the side of the dip tube **406** that is nearest the inner surface portion of the tank **102** can omit and/or be free of any holes. As such, the holes can be configured to direct water flow away from the closest inner surface portion

of the tank **102**.

(37) Further, each of the three holes **446** of each row **448** is positioned 90 degrees apart from each other. The holes **446** of each of the three rows **448** are aligned vertically to define three individual columns of holes **446**. A sum of cross-sectional area of each hole of the array of holes **440** is greater than a cross-sectional area of the dip tube **106**. A cross-sectional shape of the hole **446** is a circle, and diameter of the dip tube **406** may be in a range of about 18 mm to 22 mm. In an example, for a 19 mm diameter dip tube **406**, at least 12 mm diameter holes **446** may be defined at the side wall **442**. In an alternate embodiment, the cross-sectional shape of the hole **446** may be a rectangle. In certain embodiments, the cross-sectional shape of the hole **446** may be an oval, an ellipse, or any polygon shape known in the art.

(38) Referring to FIG. 6A, an enlarged view of a second end **534** of a dip tube **506** is illustrated, according to an embodiment of the present disclosure. The dip tube **506** includes a body portion **510** and an elongated portion **520** defining an array of holes **540** and the second end **534**. The elongated portion **520** is detachably coupled with the body portion **510** of the dip tube **506**. For example, the body portion **510** may define or be coupled to the first end of the dip tube **506** that is configured to couple with a water source. The elongated portion **520** includes a side wall **542** and an end wall **544** defining a closed terminal end for the dip tube **506**. The array of holes **540** is defined in the side wall **542** of the elongated portion **520**. In this embodiment, the elongated portion **520** is coaxially and vertically aligned with respect to a longitudinal axis of the body portion **510** of the dip tube **506**. In one embodiment, the elongated portion **520** may be attached to the body portion **510** using a snap fit mechanism or an interference fit. In another embodiment, the elongated portion **520** may be threadably attached to the body portion **510**. The array of holes **540** defined in the elongated portion **520** may be identical to the array of holes **140**, **240**, **440** and the array of slots **340** explained with reference to FIG. 2, FIG. 3, FIG. 5, and FIG. 4, respectively. Each hole **546** of the array of holes **540** defined in the elongated portion **520** is positioned to direct the water towards the wall **104** of the tank **102**. Further, each hole **546** of the array of holes **540** defined in the elongated portion **520** is adapted to direct the water substantially along the horizontal plane. With this arrangement of the elongated portion **520**, during service or maintenance of the system **100**, just the elongated portion **520** may be replaced with a new elongated portion instead of replacing the entire dip tube **506** if the elongated portion **520** malfunctions due to manufacturing defects or gets worn out after longer use of the system **100**. Thus, the service and maintenance of the system **100** becomes more efficient and cost effective.

(39) Referring to FIG. 6B, an enlarged view of a second end **634** of a dip tube **606** is illustrated, according to an embodiment of the present disclosure. The dip tube **606** includes a body portion **610** and an elongated portion **620** defining an array of holes **640** and the second end **634**. The elongated portion **620** includes a side wall **642** and an end wall **644** defining a closed terminal end for the dip tube **606**. In one embodiment, the elongated portion **620** may be detachably coupled with the body portion **610** of the dip tube **606**. In another embodiment, the elongated portion **620** may be integral with the body portion **610** to define a single body for the dip tube **606**. The body portion **610** is aligned perpendicular with respect to a longitudinal axis of the second end **634** of the dip tube **606** and parallel to the base **120** of the tank **102**. As shown in FIG. 6B, the elongated portion **620** may include a bend or elbow section between the body portion **610** and the second end **634** of the dip tube **606**.

(40) The array of holes **640** is defined in the side wall **642** of the elongated portion **620**. The array of holes **640** includes multiple rows of holes **648** radially defined around a longitudinal axis of the second end **634** of the elongated portion **620**. Each row **648** may be defined along a plane perpendicular to the longitudinal axis of the second end **634** of the elongated portion **620**. Each row **648** includes two holes **646** positioned at 180 degrees apart from each other. In another example, the two holes **646** of each row **648** may be positioned at 160 to 180 degrees apart from each other such that the holes **646** may be aligned tangential along a periphery of the side wall **642** of the dip

tube **606**. Further, the holes **646** of each row **648** are aligned vertically and thereby define two individual columns of the holes **646**, and thus a symmetrical orientation of the holes **646** is achieved at the second end **634** of the dip tube **606**. A cross-sectional shape of the hole **646** is a circle. In an alternate embodiment, the cross-sectional shape of the hole **646** may be a rectangle. In certain embodiments, the cross-sectional shape of the hole **646** may be an oval, an ellipse, or any polygon shape known in the art. Further, the array of holes **640** defined in the elongated portion **620** may be identical to the array of holes **140**, **240**, **440** and the array of slots **340** explained with reference to FIG. 2, FIG. 3, FIG. 5, and FIG. 4, respectively. Each hole **646** of the array of holes **640** defined in the elongated portion **620** is positioned to direct the water towards the wall **104** of the tank **102**. Further, each hole **646** of the array of holes **640** defined in the elongated portion **620** is adapted to direct the water substantially along the horizontal plane.

(41) Referring to FIG. 6C, an enlarged view of a second end **734** of a dip tube **706** is illustrated, according to an embodiment of the present disclosure. The dip tube **706** includes a body portion **710** and an elongated portion **720** defining an array of holes **740** and the second end **734**. The elongated portion **720** includes a side wall **742** and an end wall **744** defining a closed terminal end for the dip tube **706**. In one embodiment, the elongated portion **720** may be detachably coupled with the body portion **710**. In another embodiment, the elongated portion **720** may be integral with the body portion **710** to define a single body for the dip tube **706**. The elongated portion **720** is coaxially and vertically aligned with respect to a longitudinal axis of the body portion **710**. The elongated portion **720** has a cross sectional area that tapers towards the base **120** of the tank **102**. Particularly, for an elongated portion having a circular cross-section, a diameter defined by the side wall **742** may progressively decrease towards the end wall **744**. The array of holes **740** is defined in the side wall **742** of the elongated portion **720**. The array of holes **740** defined in the elongated portion **720** may be identical to the array of holes **140**, **240**, **440** and the array of slots **340** explained with reference to FIG. 2, FIG. 3, FIG. 5, and FIG. 4, respectively. Each hole **746** of the array of holes **740** defined in the elongated portion **720** is positioned to direct the water towards the wall **104** of the tank **102**. Further, each hole **746** of the array of holes **740** defined in the elongated portion **720** is adapted to direct the water substantially along the horizontal plane.

(42) While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed methods without departing from the spirit and scope of what is disclosed. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A water heating system comprising: a tank; one or more heat sources in thermal communication with the tank and configured to heat water within the tank; a water outlet configured to allow egress of water from the tank; and a dip tube configured to allow ingress of water into the tank, comprising: a body portion defining a first end configured to couple with a water source; and an elongated portion defining a second end distal to the first end and disposed proximate a base of the tank and configured to discharge the water to the tank substantially along a horizontal plane through an array of circular holes, the second end comprising an end cap configured to terminally close the second end and prevent vertical discharge of the water therefrom; wherein the array of holes comprises three rows of holes defined around a longitudinal axis of the dip tube, wherein each row of holes comprises three holes and is radially offset 60° from each adjacent row of holes; and wherein the body portion is axially and vertically aligned with respect to the longitudinal axis of the elongated portion and the second end of the dip tube.

2. The water heating system of claim 1, wherein each of the three holes within each row of holes

are defined at equilateral distances about a circumference of the second end of the dip tube.

3. The water heating system of claim 2, wherein the rows of holes are evenly spaced.

4. The water heating system of claim 3, wherein the adjacent holes of each row are positioned 120° apart.

5. The water heating system of claim 1, wherein each hole of the second row is aligned at an approximate middle of two adjacent holes of the first row.

6. The water heating system of claim 1, wherein the dip tube comprises one or more guide members disposed at an inner surface of a side wall thereof and configured to guide discharge of the water through the array of holes.

7. The water heating system of claim 1, wherein the elongated portion is detachably coupled with the body portion of the dip tube.

8. The water heating system of claim 1, wherein each hole of the array of holes defined in the elongated portion is positioned to direct water away from a wall of the tank.

9. The water heating system of claim 1, wherein a cross sectional area of the elongated portion tapers towards the base of the tank.

10. The water heating system of claim 1, wherein a sum of cross-sectional area of each hole of the array of holes is greater than a cross-sectional area of the second end of the dip tube.

11. The water heating system of claim 1, wherein a sum of cross-sectional area of each hole of the array of holes is at least 50 percent higher than a cross-sectional area of the second end of the dip tube.

12. The water heating system of claim 1, wherein the dip tube includes a conical protrusion extending inwardly from the second end.

13. The water heating system of claim 1, wherein a side portion of a side wall of the dip tube nearest an inner surface of the tank omits any of the array of holes.

14. The water heating system of claim 1, wherein the dip tube is configured to discharge the water such that a first hour rating of the water heater system with the dip tube is from about 1 percent to about 10 percent greater than a first hour rating of a water heater system having an otherwise identical dip tube with an open second end and no holes.
